METRIC

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27 July 2021

SUPERSEDING
MIL-DTL-24607B(SH)
w/INT. AMENDMENT 2
10 September 2013
MIL-DTL-24607B
w/AMENDMENT 1
12 October 2010

DETAIL SPECIFICATION

ENAMEL, INTERIOR, NONFLAMING (DRY), CHLORINATED ALKYD RESIN

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers enamel that has been formulated to comply with air pollution regulations which allow a maximum volatile organic compound (VOC) content of 250 grams of solvent minus water per liter of paint (2.08 pounds of solvent minus water per gallon of paint) as delivered. This specification covers chlorinated alkyd resin interior enamels that do not support combustion or flaming when dry and are for use on primed, smooth metal, glass reinforced plastic (GRP), wood, and plastic/composite surfaces. This product is supplied in a single can and is used as supplied without thinning.

1.2 Classification. Enamel covered by this specification is chlorinated alkyd resin enamel furnished in the following colors as specified (see 6.2).

a. Beach Sand, SAE AMS-STD-595 color number 22563
b. Soft White (Formula 124), SAE AMS-STD-595 color number 27880
c. Pastel Green (Formula 125), SAE AMS-STD-595 color number 24585
d. Bulkhead Gray (Formula 126), SAE AMS-STD-595 color number 26307
e. Pastel Blue, SAE AMS-STD-595 color number 25526
f. Bright White, SAE AMS-STD-595 color number 27925
g. Insignia Blue, SAE AMS-STD-595 color number 35044

Comments, suggestions, or questions on this document should be addressed to: Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 8010

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

TT-P-645 - Paint, Alkyd Type, Formula Number 84

FEDERAL STANDARDS

FED-STD-141 - Paint, Varnish, Lacquer and Related Materials: Methods of Inspection, Sampling and Testing

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-A-22262 - Abrasive Blasting Media Ship Hull Blast Cleaning

MIL-DTL-24441/20 - Paint, Epoxy-Polyamide, Green Primer, Formula 150, Type III

MIL-DTL-24441/29 - Paint, Epoxy-Polyamide, Green Primer, Formula 150, Type IV

MIL-PRF-24596 - Coating Compounds, Nonflaming, Fire-Resistant

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation contract.

CODE OF FEDERAL REGULATIONS (CFR)

29 CFR 1910.16(b)(2)(vi) - Hazard Communication, Subpart Z

40 CFR 60, Ch. 1, Appendix A-7, Method 24 - Determination of Volatile Matter Content, Water Content, Density, Volume Solids, and Weight Solids of Surface Coatings

40 CFR 261.24 - Toxicity Characteristic

(Copies of these documents are available online at www.ecfr.gov.)

U.S. ENVIRONMENTAL PROTECTION AGENCY (EPA)

EPA Method 311 - Analysis of Hazardous Air Pollutant Compounds in Paints and Coatings by Direct Injection into a Gas Chromatograph

(Copies of this document are available online at www.epa.gov.)

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NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

S9510-AB-ATM-010 - Nuclear Powered Submarine Atmosphere Control Manual

T9070-AL-DPC-020/077-2 - NAVSEA Hazardous Material Avoidance Process

(Copies of these documents are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/ by searching for the document number without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. These documents are available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AEROSPACE INDUSTRIES ASSOCIATION (AIA)

NAS 411-1 - Hazardous Material Target List

(Copies of this document are available online at www.aia-aerospace.org.)

ASTM INTERNATIONAL

ASTM D95 - Standard Test Method of Water in Petroleum Products and Bituminous Materials by Distillation

ASTM D185 - Standard Test Methods for Coarse Particles in Pigments

ASTM D522/D522M - Standard Test Methods for Mandrel Bend Test of Attached Organic Coatings

ASTM D523 - Standard Test Method for Specular Gloss

ASTM D562 - Standard Test Method for Consistency of Paints Measuring Krebs Unit (KU) Viscosity Using a Stormer-Type Viscometer

ASTM D823 - Standard Practices for Producing Films of Uniform Thickness of Paint, Coatings, and Related Products on Test Panels

ASTM D1210 - Standard Test Method for Fineness of Dispersion of Pigment-Vehicle Systems by Hegman-Type Gage

ASTM D1296 - Standard Test Method for Odor of Volatile Solvents and Diluents

ASTM D1364 - Standard Test Method for Water in Volatile Solvents (Karl Fischer Reagent Titration Method)

ASTM D1475 - Standard Test Method for Density of Liquid Coatings, Inks, and Related Products

ASTM D1849 - Standard Test Method for Package Stability of Paint

ASTM D2244 - Standard Practice for Calculation of Color Tolerances and Color Differences from Instrumentally Measured Color Coordinates

ASTM D2369 - Standard Test Method for Volatile Content of Coatings

ASTM D2698 - Standard Test Method for Determination of the Pigment Content of Solvent-Reducible Paints by High-Speed Centrifuging

ASTM D2805 - Standard Test Method for Hiding Power of Paints by Reflectometry

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ASTM D3278	-	Standard Test Methods for Flash Point of Liquids by Small Scale Closed-Cup Apparatus
ASTM D3359	-	Standard Test Methods for Measuring Adhesion by Tape Test
ASTM D3828	-	Standard Test Methods for Flash Point by Small Scale Closed Cup Tester
ASTM D4400	-	Standard Test Method for Sag Resistance of Paints Using a Multinotch Applicator
ASTM D4541	-	Standard Test Method for Pull-Off Strength of Coatings Using Portable Adhesion Testers
ASTM D5150	-	Standard Test Method for Hiding Power of Architectural Paints Applied by Roller
ASTM D5895	-	Standard Test Methods for Evaluating Drying or Curing During Film Formation of Organic Coatings Using Mechanical Recorders
ASTM D6905	-	Standard Test Method for Impact Flexibility of Organic Coatings
ASTM E162	-	Standard Test Method for Surface Flammability of Materials Using a Radiant Heat Energy Source
ASTM E1252	-	Standard Practice for General Techniques for Obtaining Infrared Spectra for Qualitative Analysis
ASTM F718	-	Standard Specification for Shipbuilders and Marine Paints and Coatings Product/Procedure Data Sheet

(Copies of these documents are available online at www.astm.org.)

SAE INTERNATIONAL

SAE AMS-STD-595	-	Colors Used in Government Procurement
SAE AMS-STD-595/24585	-	Green, Semi-Gloss, Pastel Green
SAE AMS-STD-595/25526	-	Blue, Semi-Gloss, Pastel Blue
SAE AMS-STD-595/26307	-	Gray, Semi-Gloss, Bulkhead Gray
SAE AMS-STD-595/22563	-	Orange, Semi-Gloss, Beach Sand
SAE AMS-STD-595/27880	-	Miscellaneous, Semi-Gloss, Soft White
SAE AMS-STD-595/27925	-	Miscellaneous, Semi-Gloss, Bright White
SAE AMS-STD-595/35044	-	Blue, Flat, Insignia Blue

(Copies of these documents are available online at www.sae.org.)

SOCIETY FOR PROTECTIVE COATINGS (SSPC)

SSPC-SP 10/NACE No. 2	-	Near-White Blast Cleaning

(Copies of this document are available online at www.sspc.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Composition. Enamel shall consist of ingredients in the proportions specified in table I. Tinting pigments listed in table I may be varied as necessary to conform to color requirements. The use of alternate raw material ingredients may be proposed to the Technical Authority. However, the alternate raw materials must be evaluated by the Technical Authority and the enamel shall conform to all the requirements of this specification.

TABLE I. Color name and SAE AMS-STD-595 color number.

Ingredients [1/] (kilograms)	**Soft white 27880**	**Bulkhead gray 26307**	**Pastel green 24585**	**Bright white 27925**
Chlorinated alkyd resin [2/]	212.60	211.061	214.912	225.698
Titanium dioxide [3/]	97.440	95.880	97.627	103.156
Magnesium silicate [4/]	33.992	33.498	34.106	10.306
Aluminum trihydrate [5/]	88.877	88.233	89.843	94.080
Calcium borosilicate [6/]	44.652	44.325	45.132	47.414
Calcium metaborate [7/]	43.359	43.046	43.831	46.044
PCBTF [8/]	67.054	66.569	67.785	70.715
Paint thinner [9/]	16.125	16.012	16.302	16.470
Thixatrope [10/]	6.2913	6.241	6.355	3.665
Dispersant [11/]	1.7781	1.764	1.796	1.864
Wetting agent [12/]	1.3925	1.379	1.406	1.492
Anti-skinning agent [13/]	1.3199	1.306	1.334	1.429
Cobalt drier [14/]	0.41731	0.413	0.422	0.435
Zirconium drier [15/]	0.41277	0.408	0.417	0.435
Yellow oxide paste [16/]	0.43091	10.324	1.606	---
Lamp black paste [17/]	---	3.810	---	---
Red oxide paste [18/]	---	---	0.263	---
Phthalo green paste [19/]	---	---	0.653	---
Organic yellow paste [20/]	---	---	0.803	---
Phthalo blue paste [21/]	---	---	0.118	---
LF red paste [22/]	---	---	---	---
LF orange paste [23/]	---	---	---	---
Dioxanine pigment violet [24/]	---	---	---	---
Totals	616.14	624.27	624.71	623.20

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TABLE I. Color name and SAE AMS-STD-595 color number – Continued.

Ingredients [1/] (kilograms)	Beach sand 22563	Pastel blue 25526	Insignia blue 35044
Chlorinated alkyd resin [2/]	212.46	214.63	233.53
Titanium dioxide [3/]	96.52	97.50	10.01
Magnesium silicate [4/]	33.72	34.06	49.71
Aluminum trihydrate [5/]	88.82	89.73	101.23
Calcium borosilicate [6/]	44.62	45.07	50.83
Calcium metaborate [7/]	43.33	43.77	49.71
PCBTF [8/]	67.01	67.69	89.92
Paint thinner [9/]	16.12	16.28	18.21
Thixatrope [10/]	6.28	6.37	5.91
Dispersant [11/]	1.77	1.79	2.05
Wetting agent [12/]	1.39	1.41	1.61
Anti-skinning agent [13/]	1.32	1.33	1.49
Cobalt drier [14/]	0.42	0.42	0.49
Zirconium drier [15/]	0.41	0.42	0.87
Yellow oxide paste [16/]	10.68	1.50	---
Lamp black paste [17/]	0.17	0.29	3.11
Red oxide paste [18/]	---	---	---
Phthalo green paste [19/]	---	---	---
Organic yellow paste [20/]	---	---	---
Phthalo blue paste [21/]	0.03	1.44	---
LF red paste [22/]	0.55	---	---
LF orange paste [23/]	---	0.50	---
Dioxanine pigment violet [24/]	---	---	5.84
Totals	625.61	624.18	624.52

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TABLE I. Color name and SAE AMS-STD-595 color number – Continued.

NOTES:

1/ Formulas make approximately 100 gallons of paint.
2/ Becksol 91-169, Reichold, Inc., Research Triangle Park, NC.
3/ ASTM D476, Types I or II.
4/ Talcron MP- 12-50, Specialty Minerals, Inc., Bethlehem, PA.
5/ Aluminum trihydrate, Micral 632, J.M. Huber, Edison, NJ.
6/ CW-2230, Halox Pigments, a division of ICL Specialty Products Inc, Hammond, IN.
7/ Calcium metaborate, BuLab Flamebloc BL-381 Buckman Laboratories, Memphis, TN.
8/ Oxsol 100® Equivalent EMCO, Pleasant Prairie, WI.
9/ Methyl Amyl Ketone, EMCO, Pleasant Prairie, WI.
10/ Bentone 38 (10%) Gel, NCP Coatings, Niles, MI.
11/ Spurso, Borchers Americas, Cleveland, OH.
12/ Anti-Terra 204, Byk-Chemie USA, Wallingford, CT.
13/ Methyl ethyl ketoximine, Skino #2, Borchers Americas, Westlake, OH.
14/ Cobalt Neodecanoate (12%), Borchers Americas, Westlake, OH.
15/ Zirconium Neodecanoate (18%), Borchers Americas, Westlake, OH.
16/ 5750V, Chormaflo Technologies Corp, Ashtabula, OH.
17/ 1625V, Chormaflo Technologies Corp, Ashtabula,OH.
18/ 6080V, Chormaflo Technologies Corp, Ashtabula,OH.
19/ 5150V, Chormaflo Technologies Corp, Ashtabula,OH.
20/ 5696V, Chormaflo Technologies Corp, Ashtabula,OH.
21/ 4800V, Chormaflo Technologies Corp, Ashtabula,OH.
22/ 7949V, Chormaflo Technologies Corp, Ashtabula,OH.
23/ 6012V, Chormaflo Technologies Corp, Ashtabula,OH.
24/ Hostaperm Violet RL-EDS LV3064, Clariant, Holden, MA.

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3.2.1 Metal content. When tested as specified in 4.5.1 and 4.5.1.1, the content of each soluble metal and the total content in each system component of the coating system shall not exceed the values listed in table II and table III. Total metal content may be submitted in place of soluble metal content as long as the value is lower than what is listed in the soluble metal column.

TABLE II. Soluble metals content.

Soluble metal and its compound	Maximum, mg/L
Antimony	15
Arsenic	5
Barium (excluding barite)	100
Beryllium	0.75
Cadmium	1
Chromium (VI) compounds	1
Chromium and its chromium (III) compounds	560
Cobalt	50
Copper	25
Fluoride	180
Lead	5
Mercury	0.2
Molybdenum	350
Nickel	20
Selenium	1
Silver	5
Tantalum	100
Thallium	7
Tungsten	100
Vanadium	24
Zinc and its compounds	250

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TABLE III. Metals content of coatings.

Metal and its compound in each coating dry film	Soluble metal, maximum (mg/L)	Total content, maximum (%WT)
Antimony	15.0	0.015
Arsenic	5.0	0.005
Barium (excluding barite)	100.0	0.10
Beryllium	0.75	0.0002
Cadmium	1.0	0.0005
Chromium (VI) compounds	1.0	0.001
Chromium and its chromium (III) compounds	560.0	0.56
Cobalt 1/	50.0	0.005
Copper	25.0	0.01
Fluoride salts	180.0	0.18
Lead	5.0	0.005
Mercury	0.2	0.0002
Molybdenum	350.0	0.35
Nickel	20.0	0.02
Selenium	1.0	0.002
Silver	5.0	0.001
Tantalum	100.0	0.100
Thallium	7.0	0.007
Tungsten	100.0	0.100
Vanadium	24.0	0.01
Zinc and its compounds	250.0	0.25

NOTES:

1/ Total cobalt content may exceed 0.005 %WT (up to 0.2 %WT) only if a cobalt drier is used to effect proper drying. However, soluble cobalt content may not exceed table III requirements.

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3.2.2 Hazardous air pollutants (HAPs). The content of HAP solvents (including restricted chemicals) in the total enamel shall be not greater than the percent by weight (%WT) values listed in table IV when evaluated as specified in 4.5.2.

TABLE IV. Hazardous solvent content.

Hazardous solvent in total enamel	Maximum, %WT
Benzene [1/]	0.05
Chlorinated solvent(s), total Carbon tetrachloride Chloroform (trichloromethane) [1/] Methylene chloride (dichloromethane) [1/] Tetrachloroethylene (perchloroethylene) [1/] 1,1,1-Trichloroethane (methyl chloroform) Trichloroethylene (TCE) [1/]	0.05
Ethylbenzene [1/]	0.05
Methyl, ethyl, and butyl mono-ethers of ethylene glycol or the acetates, total (methyl, ethyl, and butyl cellosolves and acetates)	0.05
Methyl ethyl ketone (MEK)	0.05
Methyl isobutyl ketone (MIBK)	0.05
Toluene [1/]	0.05
Xylene (all forms), total [1/]	0.1

NOTE:

[1/] Chemicals that are “restricted” in accordance with T9070-AL-DPC-020/077-2 for NAVSEA.

3.3 Color. The color, when tested in accordance with 4.5.3, shall be characteristic of the pigments specified in table I and shall match the colors as specified in table I and 1.2.

3.3.1 Color deviation. The color deviation of the paint shall be obtained as specified in table I. The measured color deviation (ΔE) from the specified SAE AMS-STD-595 color card in Commission Internationale de l’Eclairage (CIE) International Commission on Illumination CIELAB (CIE L*a*b*) units shall be not greater than 1 CIELAB unit.

3.4 Toxicity. When evaluated in accordance with 4.20, the enamel shall pose no serious or high risk to the health of personnel or the environment when used for its intended purpose (see 4.20 and 6.6).

3.4.1 Prohibited materials. The coating system shall not contain any chemicals categorized as “prohibited” in accordance with T9070-AL-DPC-020/077-2 for NAVSEA programs or NAS 411-1 for all other programs.

3.5 Quantitative requirements. The enamel shall conform to the quantitative characteristics specified in table V when tested as specified in section 4.

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TABLE V. Quantitative characteristics.

Characteristic	Minimum	Maximum
Pigment content, percent by mass of enamel	49.0	53.0
Volatile content, percent by mass of enamel	---	27.0
Density (gram/liter [pound/gallon])	1,557.7 (13)	1,677.6 (14)
Total solids (percent)	75.0	---
Coarse particles (percent)	---	0.5
Water content (percent)	---	0.5
Consistency (KU)	75	105
Fineness of grind (Hegman units)	5.0	---
Drying time - to touch (hours) - dry hard (hours)	--- ---	4 8
Gloss (60°) (excluding Insignia Blue, SAE AMS-STD-595 color number 35044)	45	60
Gloss (60°) (Insignia Blue, SAE AMS-STD-595 color number 35044)	---	12
Flash point (°C [°F])	37 (100)	---
Color (ΔE)	---	1.0
Sag resistance (millimeter [mils])	0.10 (4)	---
VOC (gram/liter [pound/gallon])	---	250 (2.1)
Contrast ratio	0.90	---
Crystalline silica content, (%WT of dry film)	----	0.1

3.6 Qualitative requirements.

3.6.1 Condition in container. When tested as specified in 4.5.4.4 and ASTM D1849, a freshly opened full container of the enamel shall be free of grit, seeds, skins, lumps, putrefaction, rancidity, or livering, and shall show no more surface float (separated pigments or other ingredients on the surface of the liquid enamel), pigment settling, or caking than can be readily re-incorporated to a smooth, uniform state (see 4.5.4.4). The container shall be free of corrosion.

3.6.2 Storage stability.

3.6.2.1 Partially-full container. When tested as specified in 4.5.4.1 and ASTM D1849, a three-quarter filled, closed 225-milliliter (8-ounce) glass jar of enamel shall show no skinning at the end of 48 hours, and after aging as specified, the enamel shall show no skinning, surface float (separated pigments or other ingredients on the surface of the liquid enamel), livering, curdling, hard caking, or gummy sediment. The enamel shall mix readily to a smooth, uniform state within 5 minutes when mixed with a paddle or mechanical mixer and shall have a consistency not greater than 5 units from the original Kreb unit consistency before testing (see table V). Enamel shall meet all other requirements of this specification (see 4.5.4.1).

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3.6.2.2 Full container. When tested as specified in 4.5.4.2 and ASTM D1849, the enamel in original unopened containers shall be usable for a period of 1 year of normal warehouse storage after the date of manufacture. The enamel shall show no skinning, livering, curdling, hard caking, or gummy sediment. The enamel shall mix to a smooth, uniform state with no surface float (separated pigments or other ingredients on the surface of the liquid enamel) within 5 minutes when mixed with a paddle or mechanical mixer and shall have a consistency not greater than 15 units from the original Kreb unit consistency before testing (see table V).

3.6.3 VOC regulations. Thinning of VOC compliant paint is not allowed. This paint is to be used as supplied without thinning (see table V and 4.5.5).

3.6.4 Brushing properties. When tested in accordance with 4.5.6, the enamel shall brush satisfactorily and shall dry to a uniform film, free from seeds, runs, sags, or streaks.

3.6.5 Rolling properties. When tested in accordance with ASTM D5150, the enamel shall roll satisfactorily and shall dry to a uniform film, free from seeds, runs, sags, or streaks. The dried film shall show an even, smooth finish.

3.6.6 Spraying properties. When tested in accordance with 4.5.8, the enamel shall spray satisfactorily and shall show no running, sagging, streaking, or orange peel. The air-dried film shall show no seeding, dusting, floating, mottling, hazing, or other film defects.

3.6.7 Odor. When tested in accordance with ASTM D1296, the odor of the wet enamel and the film at any interval of drying shall be characteristic of the volatiles permitted.

3.6.8 Cross-cut adhesion test. When tested as specified in ASTM D3359, a 0.025-millimeter (nominal) (1-mil [nominal]) dry film of enamel shall achieve a minimum classification of 4B.

3.6.9 Flexibility. A film of enamel shall bend without cracking or flaking (see 4.5.9).

3.6.10 Recoating. When tested in accordance with 4.5.7, recoating of a dried film shall produce no film irregularities. The enamel shall not wrinkle or lift the first coat and shall dry to a smooth, uniform finish.

3.6.11 Appearance. When tested as specified in 4.5.3 and ASTM D2244, the enamel shall show no evidence of wrinkling or any other film defects and shall have a smooth and uniform appearance.

3.6.12 Non-flaming property. When tested as specified in 4.5.11, the enamel shall be resistant to burning. The test specimen shall show no flaming in excess of 5 seconds per ignition or dripping and have an average flame spread of index of 8 with no individual specimen having a flame spread index over 10.

3.6.13 Resistance to ignition. When tested as specified in 4.5.12, the enamel shall not exhibit flaming in excess of 5 seconds per ignition, sparking, or burning drops.

3.7 Batch specific VOC certification. Manufacturer shall prepare container chemical warning label instructions for the enamel in accordance with 29 CFR 1910.16(b)(2)(iv). To comply with NESHAP requirements, the following two statements must appear on each paint can label:

a. Certification that the paint in the container meets the NESHAP requirements for shipbuilding and ship repair.

b. Statement of the ratio of volatile content to solids expressed as grams of volatile organic hazardous air pollutants (VOHAP) per liter of solids.

3.8 Off-gassing. The enamel shall be evaluated for off-gassing in accordance with the requirements of 4.22. Based on the circumstances of use and the chemical nature of the enamel, the Navy will determine whether off-gas testing is required or if an administrative assessment is acceptable. In order to be considered acceptable for use in submarines, the enamel shall be assigned either the “Permitted” or “Limited” category (see 4.22 and 6.9).

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. When specified (see 6.2), first article inspection shall be performed on a random sample drawn from the first production batch of enamel. This inspection shall include the tests specified in table VI.

TABLE VI. First article and conformance inspections.

Test	First article	Conformance	Requirements	Verification test method
Soluble and total metals content 1/	X	---	Tables II, III, and 3.2.1	4.5.1 and 4.5.1.1
Hazardous air pollutants 1/	X	---	Table IV and 3.2.2	4.5.2
Color and Surface appearance 2/	X	X	3.3 3.6.11	4.5.3 and ASTM D2244
Toxicity	X	---	3.4	4.20, 40 CFR 261 part 24
Prohibited materials	X	---	3.4.1	4.21
Pigment content 2/	X	---	Table V	ASTM D2698
Volatile percent 2/	X	---	Table V	ASTM D2369
Density 2/	X	X	Table V	ASTM D1475
Condition in container	X	---	3.6.1	4.5.4.4 and ASTM D1849
Storage stability partially-full container	X	---	3.6.2.1	4.5.4.1 and ASTM D1849
Storage stability full container	X	---	3.6.2.2	4.5.4.2 and ASTM D1849
VOC	X	---	Table V	4.5.5
Brushing properties	X	---	3.6.4	4.5.6 and FED-STD-141 4321.3
Sag resistance 2/	X	X	Table V	ASTM D4400
Rolling properties 2/	X	---	3.6.5	ASTM D5150
Spraying properties	X	---	3.6.6	4.5.8 and FED-STD-141 4331.2
Odor 2/	X	---	3.6.7	ASTM D1296
Cross-cut adhesion tape test 2/	X	---	3.6.8	ASTM D3359
Flexibility	X	---	3.6.9	4.5.9 and ASTM D6905
Resistance to ignition	X	---	3.6.13	4.5.12
Non-flaming properties	X	---	3.6.12	4.5.11 and ASTM E162
Flash point 2/	X	X	Table V	ASTM D3278 or ASTM D3828
Water content 2/	X	---	Table V	ASTM D95 or D1364
Coarse particles 2/	X	---	Table V	ASTM D185
Consistency 2/	X	X	Table V	ASTM D562
Fineness of grind 2/	X	X	Table V	ASTM D1210

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TABLE VI. First article and conformance inspections – Continued.

Test	First article	Conformance	Requirements	Verification test method
Drying time [2/]	X	X	Table V	ASTM D5895
Recoating	X	---	3.6.10	4.5.7
60-degree specular gloss	X	---	Table V	ASTM D523
Contrast ratio [2/]	X	X	Table V	ASTM D2805
Off-gassing	X	---	3.8	4.22

NOTES:

[1/] By vender certification.

[2/] Test shall be conducted in accordance with the test method prescribed by this table.

[3/] Contrast ratio determination shall be conducted on a wet film thickness of 0.08 millimeter (3 mils).

4.3 Conformance inspection. Conformance inspection shall consist of all tests specified in table VI At a minimum, conformance inspections shall be performed on the first lot and every 19,000 liters (5,000 gallons) thereafter. A lot shall consist of all components of a coating formulation from a single uniform batch or single uniform blend of batches offered for delivery at one time.

4.3.1 Sampling for conformance testing. As a minimum, the manufacturer shall select a sample of sufficient size to permit the performance of all the inspections and tests as specified in table VI. If one or more defects are found in any sample, the entire lot represented by the sample shall be rejected. When specified (see 6.2), if a lot is rejected, the manufacturer has the option of analyzing the rejected lot for the defective characteristic(s) and adjusting ingredients within allowed ranges as a new lot that shall be inspected and tested in accordance with table VI.

4.4 Test conditions. Unless otherwise specified herein, all physical tests on the coating system shall be made at 16 to 27 °C (60 to 80 °F) and 50±10 percent relative humidity.

4.4.1 Test panels and surface preparation. Unless otherwise specified (see 6.2), test panels shall be prepared as follows:

a. Two sets of panels shall be prepared and all tests conducted with a panel of each set.

b. Panels shall be steel having nominal dimensions of 150 millimeters by 300 millimeters by 3 millimeters (6 inches by 12 inches by 0.125 inch). Panels shall be prepared for testing by abrasive blasting to a near white surface in accordance with SSPC-SP 10/NACE No. 2 to achieve a minimum 0.050-millimeter (2.0-mil) surface profile. The blasted panels shall be solvent wiped to remove any traces of oil or residue, blown dry with dry air, wrapped in plastic, and stored in either a desiccator or in other storage that will prevent rusting prior to coating. The abrasive used for blasting shall be in accordance with MIL-A-22262 and be listed on the current qualified products database (QPD).

c. One set of panels shall be primed with primer provided in accordance with TT-P-645. The other set of panels shall be primed with primer provided in accordance with MIL-DTL-24441/20 or MIL-DTL-24441/29.

d. Unless otherwise specified (see 6.2), primed panels shall be dried (cured) for 1 week at normal laboratory conditions or overnight at 40 to 50 °C (104 to 122 °F) before being topcoated.

e. Unless otherwise specified (see 6.2), test coatings shall be applied in accordance with ASTM F718.

f. The enamel shall be applied to the primed panels in two coats at a wet film thickness that will yield a dry film thickness of 0.050 to 0.075 millimeter (2 to 3 mils) per coat. Unless otherwise specified (see 6.2), dry time shall be 48 hours at 21 °C (70 °F).

g. Where glass plates are specified for the test procedure, industry standard paper charts may be substituted for the test.

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4.5 Tests.

4.5.1 Metal content. Soluble and total metal content, except tantalum and tungsten, shall be determined on pulverized, cured film of the coating system in accordance with 40 CFR 261.24(a). The test results for each metal shall meet the requirements of 3.2.1. Tantalum and tungsten soluble metal content and total metal content shall be analyzed as specified in 4.5.1.1. Calculation of individual hazardous metal content can be based on either the manufacturer's testing of batches or the supplier's data for raw materials used in the product. When specified (see 6.2), a formulation value shall be provided that shall not be exceeded when the coating system is tested in accordance with this paragraph.

4.5.1.1 Tantalum and tungsten content. The tantalum and tungsten content of the enamel shall be determined using any appropriate spectroscopy test method. The tests shall be conducted in accordance with the instrument manufacturer's directions. The test results for tantalum and tungsten shall be as specified in table II and table III.

4.5.2 HAPs content. HAPs content shall be determined in accordance with EPA Method 311. Solvent fractions shall be identified in accordance with ASTM E1252. Test results shall be reported as percent by weight of the total enamel. Results shall be in accordance with the requirements of 3.2.2.

4.5.3 Color testing. Test specimens shall be prepared in accordance with ASTM D823, methods C or E using a nominal 6-mil blade film applicator onto a clear plate glass or a standard black and white Leneta chart. The color deviation of the test specimen from the SAE AMS-STD-595 color card shall be determined in accordance with ASTM D2244 using an instrument having a D_{65} light source, a 45-degree illumination angle, and a 0-degree viewing angle. The instrument shall be demonstrated to read the color of National Institute of Standards and Technology (NIST, formerly National Bureau of Standards [NBS]) traceable standards. Results shall be in accordance with the requirements of 3.3.1.

4.5.4 Storage stability.

4.5.4.1 Partially-full containers. Skinning shall be determined after 48 hours in accordance with ASTM D1849, except that a ¾-filled ½-pint, multiple friction top can shall be used. The can shall be resealed and aged for 7 days at 60 °C (140 °F). Results shall be in accordance with the requirements of 3.6.2.1.

4.5.4.2 Full container. A full quart can of enamel shall be allowed to stand undisturbed for 12 months and the contents then tested in accordance with ASTM D1849. The contents shall be evaluated for pigment settling or caking as specified in 3.6.1, and then the can shall be mixed for 5 minutes using a paddle or mechanical mixer (paint shaker) prior to re-examination. Results shall conform to the requirements of 3.6.2.2.

4.5.4.3 Accelerated storage stability. After exposure to a temperature of 60 °C (140 °F) for a period of 30 days, a previously unopened, original container of enamel shall be examined. Results shall conform to the requirements of 3.6.2, with the exception of 3.6.12 and 3.6.13.

4.5.4.4 Condition in container. A container of the product shall be equilibrated to test laboratory conditions [nominal 23 °C (73 °F)]. Evidence of pressure or vacuum in the unopened container shall be noted. The container shall then be opened and examined in accordance with ASTM D1849 for evidence of skinning, corrosion of container interior, odor of putrefaction, rancidity or souring, gel bodies, and hard settling. The enamel shall be hand-stirred 300 stirs in 2 minutes with a spatula appropriate to the container, stirring so as to ensure uniform distribution of any settled material. Results shall be in accordance with the requirements of 3.6.1.

4.5.5 Volatile organic compounds (VOC). VOC content shall be determined in accordance with 40 CFR 60, Appendix A, method 24. Results shall be in accordance with the requirements of table V (see 3.6.3).

4.5.6 Brushing properties. The coating shall be brushed in accordance with the manufacturer's instructions to a test panel prepared in accordance with 4.4.1 and tested in accordance with method 4321.3 of FED-STD-141. Results shall be in accordance with the requirements of 3.6.4.

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4.5.7 Recoating properties. Three panels shall be primed as specified in 4.4.1 with each primer. The first panel of each set shall have a 0.051- to 0.076-millimeter (2- to 3-mil) coating (topcoat) of paint applied, manufactured in accordance with MIL-PRF-24596. The second panel of each set shall have a 0.051- to 0.076-millimeter (2- to 3-mil) topcoat of Navy formula 25A, White applied. The third panel of each set shall have a 0.051- to 0.076-millimeter (2- to 3-mil) topcoat of the test enamel applied that has been manufactured in accordance with this specification. After drying the three panels for 48 hours at 50 °C (122 °F), the panels shall be cooled to ambient laboratory conditions and a 0.051- to 0.076-millimeter (2- to 3-mil) topcoat applied of the test enamel manufactured in accordance with this specification. The enamel shall be examined for blistering, wrinkling, or peeling. The adhesion shall be checked using ASTM D4541. Results shall be in accordance with the requirements in 3.6.10.

4.5.8 Spraying properties. The enamel shall be sprayed on a steel panel to a dry film thickness of 0.023 to 0.028 millimeter (0.9 to 1.1 mils). The panel shall be observed for spraying properties in accordance with method 4331.2 of FED-STD-141. Results shall be in accordance with the requirements of 3.6.6.

4.5.9 Flexibility. Flexibility shall be determined in accordance with ASTM D522/D522M. The enamel shall be applied to a flat metal plate of approximately 31 gauge coated with a dry film thickness of 0.025 millimeter (1 mil + 0.0003 inch) of TT-P-645 primer by means of a doctor blade capable of yielding a dry film thickness of 0.025 millimeter (1 mil + 0.0003 inch). The panel shall be allowed to air-dry for 2 hours then baked for 24 hours at 100 to 105 °C (212 to 221 °F). The panel shall then be removed from the oven and allowed to stand for 30 minutes at laboratory temperature (nominal 23 °C [73 °F]). The panel shall then be bent over a 3.2-millimeter (⅛-inch) mandrel or a tapered mandrel where the minimum diameter is 3.2 millimeter (⅛ inch) and the film examined at the bent under a magnification of 5 diameters. Results shall be in accordance with the requirements of 3.6.9.

4.5.10 60-degree specular gloss. The thoroughly mixed enamel shall be applied to opaque, black glass panels and tested in accordance with ASTM D823. A standard black and white Leneta chart is also acceptable. A film applicator shall be used that will produce a wet film thickness of 0.075 millimeter (3 mils). The enamel shall be dried for 24 hours under normal conditions in a dust-free environment. The applied enamel shall be dry hard before gloss test. A longer cure time shall be used if required. Testing shall be as specified in table VI. Results shall be in accordance with the requirements of table V.

4.5.11 Non-flaming property. The non-flaming property of the test enamel shall be determined in accordance with ASTM E162. Results shall be in accordance with the requirements of 3.6.12.

4.5.12 Resistance to ignition. The resistance to ignition test shall be conducted in accordance with Appendix A. Results shall be in accordance with the requirements of 3.6.13.

4.6 Pigment content. The pigment content test shall be conducted in accordance with test method prescribed by table VI. Results shall be in accordance with the requirements of table V.

4.7 Volatile percent. The volatile percent test shall be conducted in accordance with test method prescribed by table VI. Results shall be in accordance with the requirements of table V.

4.8 Density. The density test shall be conducted in accordance with test method prescribed by table VI. Results shall be in accordance with the requirements of table V.

4.9 Appearance. The appearance test shall be conducted in accordance with test method prescribed by table VI. Results shall be in accordance with the requirements of 3.6.11.

4.10 Sag resistance. The sag resistance test shall be conducted in accordance with test method prescribed by table VI. Results shall conform to be in accordance with the requirements of table V.

4.11 Rolling properties. The rolling properties test shall be conducted in accordance with test method prescribed by table VI. Results shall be in accordance with the requirements of 3.6.5.

4.12 Odor. The odor test shall be conducted in accordance with test method prescribed by table VI. Results shall be in accordance with the requirements of 3.6.7.

4.13 Cross-cut adhesion tape test. The cross-cut adhesion tape test shall be conducted in accordance with test method prescribed by table VI. Results shall be in accordance with the requirements of 3.6.8.

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4.14 Flash point. The flash point test shall be conducted in accordance with test method prescribed by table VI or as approved by the Technical Authority. Results shall be in accordance with the requirements of table V.

4.15 Water content. The water content test shall be conducted in accordance with test method prescribed by table VI. Results shall be in accordance with the requirements of table V.

4.16 Coarse particles. The coarse particles test shall be conducted in accordance with test method prescribed by table VI. Results shall be in accordance with the requirements of table V.

4.17 Consistency. The consistency test shall be conducted in accordance with test method prescribed by table VI. Results shall be in accordance with the requirements of table V.

4.18 Fineness of grind. The fineness of grind test shall be conducted in accordance with test method prescribed by table VI. Results shall be in accordance with the requirements of table V.

4.19 Drying time. The drying time test shall be conducted in accordance with test method prescribed by table I. Results shall be in accordance with the requirements of table V.

4.20 Toxicity. A Health Hazard Assessment (HHA) will be conducted to ensure conformance to 3.4, as specified (see 6.2). The Navy and Marine Corps Public Health Center (NMCPHC) will evaluate the coating systems using data provided by the manufacturer/distributor to the NMCPHC (see 3.4 and 6.6).

4.21 Prohibited materials. Prohibited materials shall be verified as specified (see 3.4.1 and 6.2).

4.22 Off-gassing. The enamel shall be evaluated for off-gassing in accordance with S9510-AB-ATM-010 chapter titled “Material Control Program” (see 3.8 and 6.9). If the Navy determines that off-gas testing is required, testing shall be conducted at a test facility approved by the Technical Authority (see 3.8). The Navy will review the off-gas test results and assign a usage category. Additionally, the Navy will assign a usage category if an administrative review is conducted in lieu of off-gas testing (see 3.8).

4.23 Fire testing provisions. All fire tests specified in this document shall be conducted by an independent testing laboratory that is accredited to ISO/IEC 17025 and is acceptable to the Technical Authority. Accreditation shall be obtained from a recognized accreditation body such as American Association for Laboratory Accreditation (A2LA) (www.A2LA.org) or International Code Council’s International Accreditation Services (IAS) (www.IASonline.org). The scope of accreditation shall include specific flammability and fire tests. All other fire test provisions shall be as specified (see 6.2 and 6.10).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point’s packaging activities within the Military Service or Defense Agency, or within the military service’s system commands. Packaging data retrieval is available from the managing Military Department’s or Defense Agency’s automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory).

6.1 Intended use. This specification covers high grade, air-drying enamels made from an alkyd resin and is intended for use on primed interior surfaces. It is characterized by excellent color and gloss retention, good drying, freedom from aftertack, and good flexibility. This enamel is a non-lead, non-chromate, and non-asbestos material. The dried paint film has been tested to U.S. EPA standards and the paint debris is, as of the time of this specification, a non-hazardous waste. This enamel has been formulated to comply with air pollution regulations which allow a maximum VOC of 250 grams of solvent minus water per liter of paint (2.08 pounds/gallon) as delivered. This product is not to be thinned for all means of application. The enamel may be applied with brush, roller, or spray. This specification covers high-grade, air-drying enamels made from alkyd resin and is intended for use on primed interior surfaces aboard naval ships.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Class, grade, and composition (see 1.2).
c. Color (number of color to SAE AMS-STD-595 recommended) required (see table I).
d. When first article is required (see 3.1).
e. Requirements for an HHA (see 3.4, 4.20, and 6.8).
f. Requirement for prohibited materials verification (see 3.4.1and 4.21).
g. When rejected lots can be adjusted and resubmitted as a new lot (see 4.3.1).
h. Test panel preparation, unless other than specified (see 4.4.1).
i. Metals content (see 4.5.1).
j. Additional fire testing provisions (see 4.23 and 6.10).
k. Batch VOC certification (see 6.4).
l. Whether SDS and ASTM F718 data sheets are required with each shipment (see 6.5).
m. Packaging requirements (see 5.1 and 6.11).

6.3 Definitions. Terms and phrases used throughout this specification are defined as follows:

6.3.1 Technical Authority. The Technical Authority has the authority, responsibility, and accountability to establish, monitor, and approve technical standards, tools, and processes in conformance to higher authority policy, requirements, architectures, and standards. Contact NAVSEA 05S for assistance with determining the appropriate Technical Authority for U.S. Navy ships.

6.4 Volatile content. Contracting officers will require the manufacturer to provide the Government with a written Volatile Organic Content (VOC) certification, as described in 40 CFR 63.785(a)(2), for each batch of coating product from which any quantity of the coating product is supplied to the Government under the contract. The Government reserves the right to reject as defective any coating product for which the VOC certification has not been provided.

6.5 Safety data sheet (SDS). When specified (see 6.2), contracting officers will identify those activities requiring copies of completed Safety Data Sheets (SDS) prepared in accordance with Appendix D of 29 CFR 1910.1200. In order to obtain the SDS, federal acquisition regulation (FAR) clause 52.223-3 will be in the contract. The contracting activity should be given an SDS at the time of contract award.

6.6 Toxicity evaluation. The NMCPHC requires sufficient information to permit an HHA of the product. Upon completion of the HHA, a copy will be provided by the NMCPHC to the Technical Authority for evaluation. The HHA process is described on the NMCPHC's website, http://www.med.navy.mil/sites/nmcphc/industrial-hygiene/Pages/health-hazard-assessment.aspx.

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6.7 Toxicological product formulations. The manufacturer should have the toxicological product formulations and associated information available for review by the NMCPHC or appropriate departmental medical service to evaluate the safety of the material for the proposed use.

6.8 HHA Evaluation by the technical authority. When specified (see 6.2), approval by the Technical Authority is required to conduct an HHA (see 4.20). The Technical Authority will request the HHA, review the vendor's formula, evaluate the HHA, and provide the HHA back to the manufacturer. The manufacturer will retain the applicable files or data package showing compliance with First Article requirements to be provided upon request to paint procuring activities. To minimize requirements for separate approval action by the Technical Authority, contract ordering data should include information related to this request, whenever possible.

6.9 Material certification. Materials to be installed in submarines are to be controlled to prevent off-gassing, which contaminates the atmosphere and can result in health hazards to personnel or deleterious effects on machinery. These controls are accomplished through the Submarine Material Control Program, which is described in the Nuclear Powered Submarine Atmosphere Control Manual, NAVSEA S9510-AB-ATM-010 chapter titled "Material Control Program." Under the Submarine Material Control Program, all materials considered for use on submarines require certification and assignment of a usage category. Under the certification process, candidate materials are selected by Navy activities or contractors, and a request for certification is submitted to the Naval Sea Systems Command, SEA 05S, 1333 Isaac Hull Ave., SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to commandstandards@navy.mil. The certification request is accompanied by detailed information, including descriptions of the material, method of application, usage, and storage. A chemical analysis is conducted, which is normally accomplished through off-gas testing. The off-gas test is required to be conducted in a Government approved laboratory designated by the preparing activity. Information pertaining to this test requirement may be obtained from the Naval Sea Systems Command, SEA 05S, 1333 Isaac Hull Ave., SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to commandstandards@navy.mil. Based on the chemical analysis results, a usage category is assigned to the material defining whether, and to what extent, the material may be used on submarines.

6.10 Additional fire testing provisions. The Technical Authority reserves the right to witness the tests and perform any of the tests set forth herein where such testing is deemed necessary to assure compliance to prescribed requirements. A2LA (www.A2LA.org) or International Code Council's IAS (www.IASonline.org) maintain a full list of the approved laboratories that are accredited to ISO/IEC 17025 and provide details on how to obtain and maintain accreditation (see 6.2).

6.11 Suggested packaging requirements. Suggested packaging is contained in table VII.

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TABLE VII. Packaging.

Packaging	Recommended requirements for direct Government acquisitions
Containers	1. The components should be furnished in cans capable of holding 0.47 liter (1 pint), 0.945 liter (1 quart), 3.78 liters (1 gallon), and 18.9 liters (5 gallons). In addition, coatings may also be furnished in cartridges, burst packs, and aerosol cans as defined in the acquisition document.
	2. Multiple friction plug containers should be in accordance with PPP-C-96, type V, class 2. Interior coatings should be as specified therein. Wire handles, as specified therein, should be provided for the 1-gallon container. Closure of the properly filled and sealed cans should be as specified in the appendix thereto.
	3. All shipping containers should comply with the requirements of the Uniform Freight Classifications (UFC), the National Motor Freight Classification (NMFC), and the applicable requirements of 49 CFR and all UN Model Regulations.
	4. Unit of procurement: the coatings covered by this specification should be purchased by volume. The unit of procurement should be in multiples of 1 liter or 1 U.S. liquid gallon at 15.5 °C (60 °F). Volumes less than 1 liter or 1 U.S. liquid gallon furnished in cartridges, packs, and aerosol cans should be purchased as defined in the acquisition document.
Packaging to ASTM D3951	1. Packaging considered commercial in nature should be in accordance with the requirements of ASTM D3951.
	2. All containers should comply with the requirements of the UFC, the NMFC, and the applicable requirements of 49 CFR and all UN Modal Regulations.
Level of pack (Packaging should be specified as follows)	1. Overseas delivery packaging. Intermediate containers of like-size coating material should be packed in close-fitting wood boxes in accordance with ASTM D6251/D6251M, overseas type, or ASTM D6880/D6880M, class 2. Box closure and strapping should be as specified in the applicable box specification or the appendix thereto, except that strapping should be flat and the finish B.
	2. Domestic delivery (level B) packing. Level B packing should be as for level A, except that boxes should be domestic type or class and the strapping should be finish A or B.
Palletization	Intermediate containers should be palletized in accordance with MIL-STD-147.
Packing for acquisitions involving direct delivery to Navy ships or installations	Treated lumber and plywood. All lumber and plywood, including laminated veneer materials, used in shipping container and pallet construction, member, blocking, bracing, and reinforcing must be fire-retardant treated material in accordance with MIL-L-19140 as follows: a. General use, weather resistant: MIL-L-19140, type II, category I. b. General use, non-weather resistant: MIL-L-19140, type I, category I.
Units	The coatings covered by this specification should be packed and packaged in cans, cartridges, and aerosol cans as defined in the acquisition document.
Safety data sheets (SDS) and product/procedure data sheets	A copy of the SDS and company product data/procedure sheets should be attached to the shipping document for each destination (see 6.5).

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TABLE VII. Packaging – Continued.

Packaging	Recommended requirements for direct Government acquisitions
Hazardous warnings	1. Labels should be in accordance with 29 CFR Parts 1910, 1915, 1917, 1918, 1926, and 1928, as well as 49 CFR.
	2. All individual containers should have the following marking (except type VI): "CAUTION: This material contains volatile solvents, with probable hazardous vapors. Use with adequate ventilation. Avoid prolonged breathing of vapors or spray mists. The solvents are highly flammable; avoid open flame and smoking."
	3. Each component container, shipping container, and palletized load should be marked with the appropriate hazardous symbol.
	4. Unit containers should be marked: "This product is asbestos, lead, hexavalent chromium, and cadmium free, except for possible trace levels."
Volatile organic content (VOC)	VOC should be stated on the label using the following statement: "Contains (insert VOC content) grams per liter (insert VOC content in pound/gallon) of volatile organic content per 40 CFR 60, Appendix A-7, method 24."
Hazardous air pollutants (HAPs)	HAPs should be stated on the label using the following statement: "Contains (insert HAP content here in gallon/liter and pound/gallon) solids (nonvolatiles) per 40 CFR 63."
Shelf life	Labels should include the product's manufacturer date and expiration date and be in accordance with MIL-STD-129.
Application instructions	Manufacturers should validate the material conforms to ASTM F718 when applicable.

6.12 Subject term (key word) listing.

Asbestos

Cobalt compounds

Paint

Semigloss

Topcoat

Volatile organic content

6.13 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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RESISTANCE TO IGNITION

A.1. SCOPE

A.1.1 Scope. This procedure determines whether a paint will ignite into flames when the sole ignition source is the heated metallic substrate to which the paint has been applied. This Appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

A.1.2 Resistance to auto-ignition. The test yields information on the ability of the coating to resist auto-ignition, running, and dripping when subjected to a very rapid heat rise and information on the ability of the paint to resist spread of the flame across the test specimen, as well as the tendency of a paint to ignite secondary flames by flaming drips.

A.1.3 Test conditions. The conditions of the test have been standardized so that paints passing this test may be safely used on ships' bulkheads and overheads where non-flaming paints are required to prevent fire from spreading from one sealed compartment to the next during a severe shipboard fire.

A.2. TEST APPARATUS

A.2.1 Test apparatus. The recommended apparatus consists of a high amperage direct current (DC) generator capable of 300 to 400 amperes, electrodes for holding the specimen, an ammeter attached across an in-line shunt, a voltmeter, and a switch capable of handling the voltages and currents involved. The wiring diagram is shown on figure A-1. The current from the DC generator is shunted directly through the test specimen to cause rapid heating. 3/0 stranded welding cable is recommended for wiring.

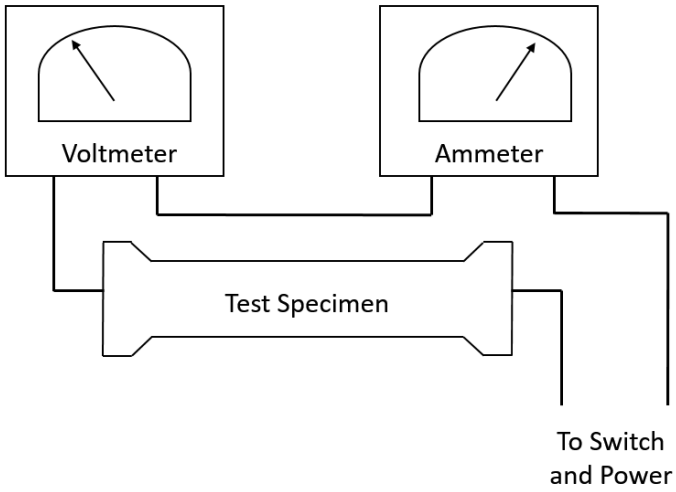


FIGURE A-1. Wiring diagram.

A.2.1.1 Test apparatus location. The test apparatus shall be arranged in a shielded cabinet or room to reduce air current effects. Air moving past the specimen shall be regulated to that caused only by convection heating of the test specimen.

A.2.2 Test panels preparation. Test specimens shall be coated to either 0.15 millimeter (6 mils) dry film thickness in as many coats as necessary to achieve 0.15 millimeter (6 mils), or to that thickness recommended by the manufacturer's ASTM F718, whichever is greater. If multiple coats are required to achieve the test thickness, at least 24 hours, but no more than 48 hours shall lapse between coats. Fully coated test specimens shall be aged 14 days before being tested.

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A.3. PROCEDURE

A.3.1 Calibration of temperature. Calibration of the 1,000 °C (1,832 °F) in 60 seconds time – temperature curves for the particular test specimens and power source shall be conducted. The time-temperature curve is determined by the cross section and resistivity of the actual test specimens used. Temperature-indicating crayons, temperature-indicating paints, or optical pyrometers are recommended. Thermocouples are not recommended due to the current flowing in the test specimen.

A.3.2 Test specimen observation. The test specimen shall be observed, and the following shall be recorded:

a. Auto ignition. Flaming (ignition) of the paint before reaching 1,000 °C (1,832 °F).

b. Smoke generation. Fumes or smoke from the paint and the time at which fumes/smoke began.

c. Smoke observation (qualitative). Volume of smoke (light, moderate, or heavy).

d. Performance observations. Other characteristics, such as slouching, dripping, softening, or color changes.

e. Documentation. Video recording and color slides shall be taken during and after the test, particularly at the time of ignition and at the point of maximum involvement, and after the completion of the test.

A.3.3 Analysis of test results. Test results shall be in accordance with the requirements of 3.6.13.

A.4. SUMMARY

A.4.1 Summary of test method and results. This test procedure provides a relatively simple, straightforward, and reproducible method to determine the ignition potential of materials used to paint the interior surfaces of a compartment. In addition, important data on the reaction of the paint to heating of the substrate is determined. The fact that data is a “go” or “no-go” form makes the error of operation relatively small.

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CONCLUDING MATERIAL

Custodians:
Army – CR4
Navy – SH

Preparing activity:
Navy – SH
(Project 8010-2020-008)

Review activities:
Army – MR
Navy – CG, MC, YD

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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